

MODULE 1 | PART 2
POLYNOMIALS
IOQM & RMO 2026

POLYNOMIALS

Basics of Polynomials

Definitions, Notation, Terminology

- ❖ A polynomial in variable x is an expression $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$, with coefficients $a_i \in \mathbb{R}$.
- ❖ Degree: the highest power n with $a_n \neq 0$.
- ❖ Terms: each monomial $a_i x^i$.
- ❖ Leading coefficient: a_n .
- ❖ Constant term: a_0 .
- ❖ Standard form: written in descending powers of x .
- ❖ Zero (root): value r such that $p(r) = 0$.

Example:

Identify the degree, leading coefficient, and constant term of

$$p(x) = 4x^5 - 3x^2 + 7x - 9$$

Solution:

Degree = 5, leading coefficient = 4,
constant term = -9.

Example:

Evaluate the polynomial

$$q(x) = 2x^3 - 5x + 1$$

at $x = 2$ and $x = -1$.

Solution:

At $x = 2$: $2 \cdot 8 - 5 \cdot 2 + 1 = 16 - 10 + 1 = 7$.

At $x = -1$: $2(-1)^3 - 5(-1) + 1 = 4$

Example:

Add and simplify:

$$(3x^2 + x - 4) + (5x^2 - 2x + 7)$$

Solution:

Combine like terms:

$$\begin{aligned}(3x^2 + 5x^2) + (x - 2x) + (-4 + 7) \\ = 8x^2 - x + 3.\end{aligned}$$



Example:

Multiply and simplify:

$$(x - 3)(2x^2 + x - 5)$$

Solution:

Distribute:

$$\begin{aligned} x \cdot (2x^2 + x - 5) - 3 \cdot (2x^2 + x - 5) \\ = (2x^3 + x^2 - 5x) - (6x^2 + 3x - 15) \\ = 2x^3 + x^2 - 5x - 6x^2 - 3x + 15 \\ = 2x^3 - 5x^2 - 8x + 15 \end{aligned}$$

Example:

Classify each polynomial by degree:

- (i) $7x - 2$,
- (ii) $4x^4 + x^2 + 1$,
- (iii) $6x^3 - x + 5$.

Solution:

- (i) linear (degree 1),
- (ii) quartic (degree 4),
- (iii) cubic (degree 3).

Example:

Find the zeros of the polynomial

$$r(x) = 5x^2(x - 2)(x + 3)$$

Solution:

Zeros when any factor = 0:

$$\begin{aligned} 5x^2 = 0 &\implies x = 0, \\ x - 2 = 0 &\implies x = 2, \\ x + 3 = 0 &\implies x = -3 \\ x &= \{0, 2, -3\} \end{aligned}$$

Linear Polynomials

❖ A linear polynomial in x has the form $p(x) = ax + b, a \neq 0$.

❖ Slope-Intercept Form: $y = ax + b$

- Slope = a
- y -intercept = $(0, b)$

❖ Zero (root or x -intercept):

Solve $ax + b = 0$

$$x = -\frac{b}{a}, \quad \text{point} \left(-\frac{b}{a}, 0 \right).$$

❖ Graph: a straight line through $(0, b)$ with steepness a .

❖ To plot: mark the y -intercept, then $\frac{\text{rise}}{\text{run}} = a$.

Example:

For $p(x) = 3x - 2$, find the slope, the y -intercept and the x -intercept.

Solution:

- Slope $a = 3$.
- y -intercept at $(0, -2)$.
- x -intercept: solve $3x - 2 = 0$
 $\implies x = \frac{2}{3}, \text{ so } \left(\frac{2}{3}, 0 \right).$



Example:

Find the zero of $q(x) = -4x + 8$.

Solution:

Solve $-4x + 8 = 0$:

$$-4x = -8 \implies x = 2$$

Zero at $(2, 0)$.

Quadratic Polynomials

- ❖ A quadratic polynomial has the form $p(x) = ax^2 + bx + c$ with $a \neq 0$.
- ❖ Its graph is a parabola opening upward if $a > 0$, downward if $a < 0$.
- ❖ Vertex: $\left(-\frac{b}{2a}, p\left(-\frac{b}{2a}\right)\right)$.
- ❖ Axis of symmetry: $x = -\frac{b}{2a}$.
- ❖ Discriminant: $\Delta = b^2 - 4ac$:
 - $\Delta > 0$: two distinct real roots.
 - $\Delta = 0$: one (double) real root.
 - $\Delta < 0$: no real roots.
- ❖ Factored form: $a(x - r_1)(x - r_2)$ when roots r_1, r_2 exist.

**Example:**

For $p(x) = 2x^2 - 8x + 3$:

- (i) Find the vertex.
- (ii) Determine the axis of symmetry.
- (iii) State whether the parabola opens up or down.

Solution:

$$x_v = -\frac{-8}{2 \cdot 2} = 2,$$

$$p(2) = 2 \cdot 4 - 16 + 3 = -5.$$

So vertex $(2, -5)$.

Axis: $x = 2$.

Since $a = 2 > 0$, it opens upward.

Example:

Solve the quadratic equation:

$$3x^2 - 5x + 2 = 0$$

Solution:

$$\begin{aligned} \text{Discriminant: } \Delta &= (-5)^2 - 4 \cdot 3 \cdot 2 \\ &= 25 - 24 = 1. \end{aligned}$$

$$\text{Roots: } x = \frac{5 \pm 1}{2 \cdot 3} = \frac{5 \pm 1}{6} \implies x = 1, \frac{2}{3}$$



Example:

Factor and solve:

$$x^2 + 5x + 6 = 0$$

Solution:

We seek two numbers whose sum is 5 and product is 6 : 2 and 3.

$$\text{So } x^2 + 5x + 6 = (x + 2)(x + 3) = 0$$

$$\implies x = -2, -3$$

Example:

Given roots satisfy $r_1 + r_2 = 4$ and $r_1 r_2 = 3$, write the monic quadratic.

Solution:

A monic quadratic with roots r_1, r_2 is

$$x^2 - (r_1 + r_2)x + r_1 r_2 = x^2 - 4x + 3$$

Example:

For $q(x) = -x^2 + 4x + 5$:

(i) Find its x -intercepts.

(ii) Find its orientation and vertex.

Solution:

$$-x^2 + 4x + 5 = 0 \iff x^2 - 4x - 5 = 0$$

$$x = \frac{4 \pm \sqrt{16 + 20}}{2} = \frac{4 \pm 6}{2} \implies x = 5, -1$$

$$\text{Vertex at } x_v = -\frac{4}{2(-1)} = 2,$$

$$q(2) = -4 + 8 + 5 = 9.$$

Opens downward ($a = -1$).

Intercepts at $(-1, 0)$ and $(5, 0)$, vertex $(2, 9)$.

Example:

Find the values of k for which

$$x^2 - 6x + k = 0$$

has a double root.

Solution:

Double root when $\Delta = 0$:

$$(-6)^2 - 4 \cdot 1 \cdot k = 0 \implies 36 - 4k = 0$$

$$\implies k = 9.$$



1. *The equations $x^2 - 4x + k = 0$ and $x^2 + kx - 4 = 0$, where k is a real number, have exactly one common root. What is the value of k ?

[PRMO 2015]



2. *Let $f(x) = x^3 - 3x + b$ and $g(x) = x^2 + bx - 3$, where b is a real number. What is the sum of all possible values of b for which the equations $f(x) = 0$ and $g(x) = 0$ have a common root?

[PRMO 2013]



3. *Let a and b be the roots of the equation $x^2 + x - 3 = 0$. Find the value of the expression $4b^2 - a^3$.

[PRMO 2016]



Cubic Polynomials

- ❖ A cubic polynomial has the form $p(x) = ax^3 + bx^2 + cx + d$, $a \neq 0$.
- ❖ It can have up to 3 real roots; graph has at most two turning points.
- ❖ End-behavior: as $x \rightarrow \pm\infty$, $p(x) \rightarrow \pm\infty$ if $a > 0$ (reversed if $a < 0$).

❖ Vieta's formulas: if roots are r_1, r_2, r_3 , then

$$r_1 + r_2 + r_3 = -\frac{b}{a},$$

$$r_1r_2 + r_2r_3 + r_3r_1 = \frac{c}{a},$$

$$r_1r_2r_3 = -\frac{d}{a}$$

❖ Factor Theorem: if $p(r) = 0$, then $(x - r)$ is a factor.

❖ To solve: factor out one linear factor, then solve the remaining quadratic.

Example:

For $p(x) = 2x^3 - 3x^2 + x - 5$, find $r_1 + r_2 + r_3$, $r_1r_2 + r_2r_3 + r_3r_1$ and $r_1r_2r_3$.

Solution:

By Vieta's Formulas:

$$r_1 + r_2 + r_3 = -\frac{-3}{2} = \frac{3}{2},$$

$$r_1r_2 + r_2r_3 + r_3r_1 = \frac{1}{2},$$

$$r_1r_2r_3 = -\frac{-5}{2} = \frac{5}{2}$$

Example:

Solve the equation:

$$x^3 + x^2 - x - 1 = 0$$

Solution:

Factor by grouping:

$$\begin{aligned} x^2(x+1) - 1(x+1) &= (x+1)(x^2-1) \\ &= (x+1)^2(x-1) \end{aligned}$$

Roots: $x = -1$ (double), $x = 1$.

Example:

Find the monic cubic with roots 1, 2, -3.

Solution:

If roots are 1, 2, -3, the monic polynomial is

$$\begin{aligned} & (x-1)(x-2)(x+3) \\ &= (x^2 - 3x + 2)(x+3) \\ &= x^3 + 0x^2 - 7x + 6 \end{aligned}$$

Example:

Determine the zeros and their multiplicities of

$$(2x-1)(x+3)(x-2) = 0$$

Solution:

Set each factor equal to zero:

$$2x-1=0 \implies x = \frac{1}{2},$$

$$x+3=0 \implies x = -3,$$

$$x-2=0 \implies x = 2.$$

Each is simple (multiplicity 1).

n^{th} Degree Polynomials

- ❖ A general polynomial of degree n has the form

$$\begin{aligned} p(x) &= a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0, \\ a_n &\neq 0 \end{aligned}$$

- ❖ Fundamental Theorem of Algebra: $p(x)$ has exactly n complex roots (counting multiplicity).
- ❖ End-behaviour: as $x \rightarrow \pm\infty$,
 $p(x) \sim a_n x^n$.

Example:

For $p(x) = -3x^8 + x^5 - 6x^2 + 4$, identify:

- degree, leading coefficient, constant term,
- end-behaviour as $x \rightarrow \pm\infty$.

Solution:

- Degree $n = 8$,
leading coefficient $a_8 = -3$,
constant term $a_0 = 4$.
- As $x \rightarrow \infty$, $-3x^8 \rightarrow -\infty$.
As $x \rightarrow -\infty$, $-3x^8 \rightarrow -\infty$
(even power).

Example:

Given

$$q(x) = x^4 - 2x^3 + 3x^2 - x + 5$$

find the sum of its roots.

Solution:

By Vieta's Formula:

$$r_1 + r_2 + r_3 + r_4 = -\frac{-2}{1} = 2$$

Example:

Find the product of the roots of

$$2x^5 - 3x^4 + 5x^2 - 6 = 0$$

Solution:

For $p(x) = 2x^5 - 3x^4 + 0x^3 + 5x^2 + 0x - 6$

$$\prod_{i=1}^5 r_i = (-1)^5 \frac{a_0}{a_5} = -\frac{-6}{2} = 3.$$

Example:

Construct the monic quartic polynomial with roots

$$1, -1, 2, -2.$$

Solution:

Roots are $\pm 1, \pm 2$, so the required polynomial is

$$\begin{aligned} & (x-1)(x+1)(x-2)(x+2) \\ &= (x^2-1)(x^2-4) = x^4 - 5x^2 + 4. \end{aligned}$$

Example:

If the roots of $p(x) = x^3 + ax^2 + bx + 1$ are r_1, r_2, r_3 , find the polynomial whose roots are $\frac{1}{r_1}, \frac{1}{r_2}, \frac{1}{r_3}$.

Solution:

The reciprocal-root polynomial is

$$\begin{aligned} x^3 p\left(\frac{1}{x}\right) &= x^3 \left(\frac{1}{x^3} + a \frac{1}{x^2} + b \frac{1}{x} + 1 \right) \\ &= 1 + ax + bx^2 + x^3 \end{aligned}$$

Hence $\boxed{x^3 + bx^2 + ax + 1}$.



LCM and HCF of Polynomials

- ❖ Highest Common Factor (HCF) of $f(x), g(x)$ is the monic polynomial of greatest degree that divides both.
- ❖ Least Common Multiple (LCM) of $f(x), g(x)$ is the monic polynomial of least degree divisible by both.
- ❖ Relation: for nonzero f, g ,

$$\text{HCF}(f, g) \text{LCM}(f, g) = f(x)g(x),$$
 up to a constant factor.

Example:

Find the HCF and LCM of

$$f(x) = x^2 - 1, g(x) = x^2 - 3x + 2$$

Solution:

Factor:

$$f(x) = (x-1)(x+1)$$

$$g(x) = (x-1)(x-2)$$

Common factor $(x-1)$.

HCF is $\boxed{x-1}$.

$$\begin{aligned} \text{LCM} &= \frac{f(x)g(x)}{\text{HCF}} \\ &= \frac{(x-1)(x+1)(x-1)(x-2)}{(x-1)} \end{aligned}$$

$$= (x-1)(x+1)(x-2)$$

LCM is $\boxed{(x-1)(x+1)(x-2)}$.





4. *Suppose 1, 2, 3 are the roots of the equation $x^4 + ax^2 + bx = c$. Find the value of c .

[PRMO 2017]



Partial Fractions

Rational Expressions & Partial Fractions

- ❖ A rational expression is a quotient $\frac{P(x)}{Q(x)}$ of polynomials, $Q \neq 0$.
- ❖ Domain: all real x for which $Q(x) \neq 0$.
- ❖ Simplification: factor numerator and denominator and cancel common factors.
- ❖ A rational expression is proper if $\deg P < \deg Q$, else improper.
- ❖ Partial-fraction decomposition: express a proper rational expression as a sum of simpler terms over linear and irreducible quadratic factors:

$$\frac{P}{(x-a)^r} + \frac{A}{(x-a)^k} + \frac{Bx+C}{x^2+bx+c} + \dots$$

Example:

Simplify: $\frac{x^2 - 1}{x - 1}$.

Solution:

Factor numerator:

$$\frac{(x-1)(x+1)}{x-1} = x+1$$

valid for $x \neq 1$

Example:

Decompose into partial fractions: $\frac{1}{x^2 - 1}$.

Solution:

Write

$$\frac{1}{(x-1)(x+1)} = \frac{A}{x-1} + \frac{B}{x+1}$$

Clear denominators:

$$1 = A(x+1) + B(x-1)$$

$$\text{Set } x = 1: 1 = 2A \implies A = \frac{1}{2}$$

$$\text{Set } x = -1: 1 = -2B \implies B = -\frac{1}{2}$$

$$\text{Thus } \frac{1}{x^2 - 1} = \frac{1/2}{x-1} - \frac{1/2}{x+1}$$



Example:

Decompose: $\frac{2x+3}{x^2-x-2}$.

Solution:

Factor denominator:

$x^2 - x - 2 = (x-2)(x+1)$. Write

$$\frac{2x+3}{(x-2)(x+1)} = \frac{A}{x-2} + \frac{B}{x+1}$$

Clear denominators:

$$2x+3 = A(x+1) + B(x-2)$$

$$\text{Set } x = 2: 7 = 3A \implies A = \frac{7}{3}.$$

$$\text{Set } x = -1: 1 = -3B \implies B = -\frac{1}{3}.$$

$$\text{So } \frac{2x+3}{x^2-x-2} = \frac{7/3}{x-2} - \frac{1/3}{x+1}$$

Example:

Decompose: $\frac{3+x}{(x-1)^2(x+2)}$.

Solution:

$$\text{Set } \frac{x+3}{(x-1)^2(x+2)} = \frac{A}{(x-1)} + \frac{B}{(x-1)^2} + \frac{C}{(x+2)}$$

Clear denominators:

$$x+3 = A(x-1)(x+2) + B(x+2) + C(x-1)^2$$

Expand and equate coefficients:

$$x+3 = A(x^2+x-2) + B(x+2) + C(x^2-2x+1)$$

$$\text{Collect } x^2: 0 = A + C.$$

$$\text{Collect } x: 1 = A + B - 2C.$$

$$\text{Constant: } 3 = -2A + 2B + C.$$

Solve:

$$C = -A,$$

$$1 = A + B + 2A = 3A + B,$$

$$3 = -2A + 2B - A = -3A + 2B.$$

$$\implies A = \frac{-1}{9}, B = \frac{4}{3}, C = \frac{1}{9}$$

$$\text{So } \frac{x+3}{(x-1)^2(x+2)} = -\frac{1}{9(x-1)} + \frac{4}{3(x-1)^2} + \frac{1}{9(x+2)}$$

Polynomial Division

- ❖ Given polynomials $f(x)$ and $g(x) \neq 0$, there exist unique $Q(x)$ and $R(x)$ such that

$$f(x) = g(x)Q(x) + R(x),$$

where $\deg R < \deg g$

- ❖ Long Division: Align terms, divide leading terms, multiply back, subtract, bring down next term, repeat.

Example:

Divide: $\frac{x^3 - 6x^2 + 11x - 6}{x - 1}$.

Solution:

By synthetic division with $a = 1$:

1	1	-6	11	-6
		1	-5	6
	1	-5	6	0

Quotient $Q(x) = x^2 - 5x + 6$, remainder 0.

$$Q(x) = x^2 - 5x + 6, R = 0.$$

Example:

Divide: $\frac{2x^4 + 3x^3 - x + 5}{x^2 + x - 2}$

Solution:

Long-division steps:

1. $2x^4 / (x^2) = 2x^2$.

2. Multiply divisor:

$$2x^2 (x^2 + x - 2) = 2x^4 + 2x^3 - 4x^2.$$

3. Subtract:

$$\begin{aligned} (2x^4 + 3x^3 + 0x^2 - x + 5) \\ - (2x^4 + 2x^3 - 4x^2) \\ = x^3 + 4x^2 - x + 5. \end{aligned}$$

4. Next term: $x^3 / (x^2) = x$

5. Multiply: $x (x^2 + x - 2) = x^3 + x^2 - 2x$

6. Subtract:

$$\begin{aligned} (x^3 + 4x^2 - x + 5) - (x^3 + x^2 - 2x) \\ = 3x^2 + x + 5 \end{aligned}$$

7. Next: $3x^2 / (x^2) = 3$.

8. Multiply: $3 (x^2 + x - 2)$

$$= 3x^2 + 3x - 6$$

9. Subtract:

$$\begin{aligned} (3x^2 + x + 5) - (3x^2 + 3x - 6) \\ = -2x + 11 \end{aligned}$$

Quotient $Q(x) = 2x^2 + x + 3$,

remainder $R(x) = -2x + 11$.

$$Q(x) = 2x^2 + x + 3, R(x) = -2x + 11$$

Example:

Divide: $\frac{x^4 - 1}{x^2 - 1}$

Solution:

Factor both:

$$\frac{(x^2 - 1)(x^2 + 1)}{x^2 - 1} = x^2 + 1, \text{ with remainder } 0.$$

$$Q(x) = x^2 + 1, R = 0.$$



5. *Let x, y be positive integers such that

$$x^4 = (x - 1)(y^3 - 23) - 1$$

Find the maximum possible value of $x + y$.

[IOQM 2023]



Remainder Theorem

- ❖ When a polynomial $f(x)$ is divided by $x - a$, the remainder is $f(a)$.
- ❖ Equivalently,
$$f(x) = (x - a)Q(x) + f(a),$$
for some quotient $Q(x)$.
- ❖ Quickly compute remainders without long division by evaluating at $x = a$.
- ❖ Useful for testing factors (if $f(a) = 0$, then $x - a$ is a factor).

Example:

Find the remainder when

$$f(x) = x^3 + 2x^2 - 5x + 3$$

is divided by $x - 1$.

Solution:

By the Remainder Theorem, remainder = $f(1)$
 $f(1) = 1 + 2 - 5 + 3 = 1$.

Example:

Find the remainder when

$$f(x) = 2x^4 - 3x^3 + 4x - 5$$

is divided by $x + 1$.

Solution:

Here $x + 1 = x - (-1)$, so evaluate $f(-1)$:

$$\begin{aligned} f(-1) &= 2 - (-3) - 4 - 5 \\ &= 2 + 3 - 4 - 5 = -4. \end{aligned}$$

Example:

Determine k so that the remainder when

$$f(x) = x^3 + kx^2 + 4x + 1$$

is divided by $x - 1$ equals 3.

Solution:

Require $f(1) = 3$:

$$\begin{aligned} 1 + k + 4 + 1 &= 3 \implies k + 6 = 3 \\ \implies k &= -3. \end{aligned}$$

Factor Theorem

- ❖ Statement: $x - a$ is a factor of the polynomial $f(x)$ if and only if $f(a) = 0$.
- ❖ Use to factor polynomials by testing likely roots (e.g., integer divisors of the constant term).
- ❖ Once one linear factor is found, divide it out to reduce degree and factor the quotient.
- ❖ Efficient for finding all real (and sometimes complex) roots of low-degree polynomials.

Example:

Show that $(x - 2)$ is a factor of

$$f(x) = x^3 - x^2 - 4x + 4$$

Solution:

Compute $f(2)$:

$$2^3 - 2^2 - 4 \cdot 2 + 4 = 8 - 4 - 8 + 4 = 0$$

Since $f(2) = 0$, by the Factor Theorem $x - 2$ is a factor.

Example:

Factor completely:

$$x^3 - 3x^2 - 4x + 12$$

Solution:

Test $x = 3$:

$$3^3 - 3 \cdot 3^2 - 4 \cdot 3 + 12 = 27 - 27 - 12 + 12 = 0$$

So $x - 3$ is a factor. Divide:

$$x^3 - 3x^2 - 4x + 12 = (x - 3)(x^2 - 4)$$

Then $x^2 - 4 = (x - 2)(x + 2)$.

Hence $(x - 3)(x - 2)(x + 2)$.

Example:

Verify that $x + 1$ is a factor and factor:

$$x^4 - x^3 - x^2 + x$$

Solution:

Compute $f(-1)$:

$$\begin{aligned} (-1)^4 - (-1)^3 - (-1)^2 + (-1) \\ = 1 + 1 - 1 - 1 = 0 \end{aligned}$$

So $x + 1$ is a factor.

Divide:

$$x^4 - x^3 - x^2 + x = (x + 1)(x^3 - 2x^2 + x)$$

Factor the cubic:

$$\begin{aligned} x^3 - 2x^2 + x &= x(x^2 - 2x + 1) \\ &= x(x - 1)^2. \end{aligned}$$

Thus $(x + 1)x(x - 1)^2$.

**Example:**

Determine k such that $(x + 2)$ is a factor of

$$f(x) = x^3 + kx^2 - x - 2$$

Solution:

Require $f(-2) = 0$:

$$(-2)^3 + k(-2)^2 - (-2) - 2$$

$$= -8 + 4k + 2 - 2 = 4k - 8 = 0$$

$$\implies k = 2.$$

Example:

Factor using the Factor Theorem:

$$x^3 + 4x^2 - x - 4$$

Solution:

Test $x = 1$: $1 + 4 - 1 - 4 = 0$.

So $x - 1$ is a factor.

Divide:

$$x^3 + 4x^2 - x - 4 = (x - 1)(x^2 + 5x + 4)$$

Factor the quadratic:

$$x^2 + 5x + 4 = (x + 1)(x + 4)$$

Hence $(x - 1)(x + 1)(x + 4)$.



Intermediate Value Theorem

Intermediate Value Theorem for Polynomials

- ❖ Polynomials are continuous functions on $\mathbb{R}$.
- ❖ Intermediate Value Theorem (IVT): If f is continuous on $[a, b]$ and $f(a)$ and $f(b)$ have opposite signs, then there exists $c \in (a, b)$ such that $f(c) = 0$.
- ❖ IVT guarantees at least one real root in any sign-change interval.
- ❖ Does not locate the root exactly—only confirms existence.
- ❖ Useful for bracketing roots before refinement by other methods.

Example:

Show that the polynomial

$$f(x) = x^3 - x - 2$$

has a real root in the interval $[1, 2]$.

Solution:

Compute $f(1) = 1 - 1 - 2 = -2$ and

$$f(2) = 8 - 2 - 2 = 4.$$

Since $f(1) < 0$ and $f(2) > 0$, by IVT there is $c \in (1, 2)$ with $f(c) = 0$.



Example:

Show that

$$g(x) = x^2 - 2$$

has a real root in the interval $[1, 2]$.

Solution:

$$g(1) = 1 - 2 = -1 < 0,$$

$$g(2) = 4 - 2 = 2 > 0.$$

By IVT there exists $c \in (1, 2)$ with $g(c) = 0$,
i.e. $\sqrt{2}$.

Example:

f is a monic polynomial of degree 5 such that $f(0) = 1, f(1) = 2, f(2) = 5, f(3) = 10$ and $f(4) = 17$. Find the value of $f(5)$.

Solution:

$$g(x) = f(x) - (x^2 + 1)$$

$$g(0) = 0, g(1) = 0, g(2) = 0, g(3) = 0$$

$$g(4) = 0$$

$$\Rightarrow g(x) = x(x-1)(x-2)(x-3)(x-4)$$

$$\Rightarrow f(x) = x(x-1)(x-2)(x-3)(x-4) + x^2 + 1$$

$$f(5) = 5! + 25 + 1 = 146$$



6. The quotient and remainder when $x^4(1-x)^4$ is divided by $1+x^2$ is



7. *Let $f(x)$ and $g(x)$ be two real polynomials of degree 2 and 1 respectively. If $f(g(x)) = 8x^2 - 2x$, and $g(f(x)) = 4x^2 + 6x + 1$, then the value of $f(2) + g(2)$ is

[JEE Main 2022]



8. *Let $f(x)$ be a polynomial of degree 3 such that $f(k) = -\frac{2}{k}$ for $k = 2, 3, 4, 5$. Then the value of $52 - 10f(10)$ is equal to

[JEE Main 2021]



9. Let $f(x)$ be a polynomial with real coefficients of degree 2. Suppose that for some pairwise distinct nonzero real numbers a, b, c we have

$$f(a) = bc; f(b) = ca; f(c) = ab$$

Determine $f(a + b + c)$ in terms of a, b, c .

[RMO 2023]



10. Let $P(x) = x^2 + \frac{1}{2}x + b$ and $Q(x) = x^2 + cx + d$ be two polynomials with real coefficients such that $P(x)Q(x) = Q(P(x))$ for all real x . Find all the real roots of $P(Q(x)) = 0$.

[RMO 2017]





11. If the roots of the equation $x^2 - 10cx - 11d = 0$ are a, b and those of $x^2 - 10ax - 11b = 0$ are c, d (where a, b, c, d are distinct numbers), then the value of $a + b + c + d$ is

[IIT JEE 2006]



12. Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 20x - 2020$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 20x + 2020$. Then the value of $ac(a - c) + ad(a - d) + bc(b - c) + bd(b - d)$ is

- (A) 0 (B) 8000
(C) 8080 (D) 16000

[JEE Adv 2020]



13. Let $f(x)$ be a quadratic polynomial such that $f(-1) + f(2) = 0$. If one of the roots of $f(x) = 0$ is 3, then its other root lies in

- (A) (1, 3) (B) (-3, -1)
(C) (0, 1) (D) (-1, 0)

[JEE Main 2020]



14. If $g(x) = x^2 + x - 1$ and $(g \circ f)(x) = 4x^2 - 10x + 5$, then $f\left(\frac{5}{4}\right)$ is equal to

- (A) $-\frac{3}{2}$ (B) $-\frac{1}{2}$
(C) $\frac{1}{2}$ (D) $\frac{3}{2}$

[JEE Main 2021]



ANSWER KEY

(1) 3

(2) 0

(3) 19

(4) 36

(5) 7

(6) $Q(x) = x^6 - 4x^5 + 5x^4 - 4x^2 + 4$

$R(x) = -4$

(7) 18

(8) 26

(9) $ab + bc + ac$

(10) $\frac{1}{2}, -1$

(11) 1210

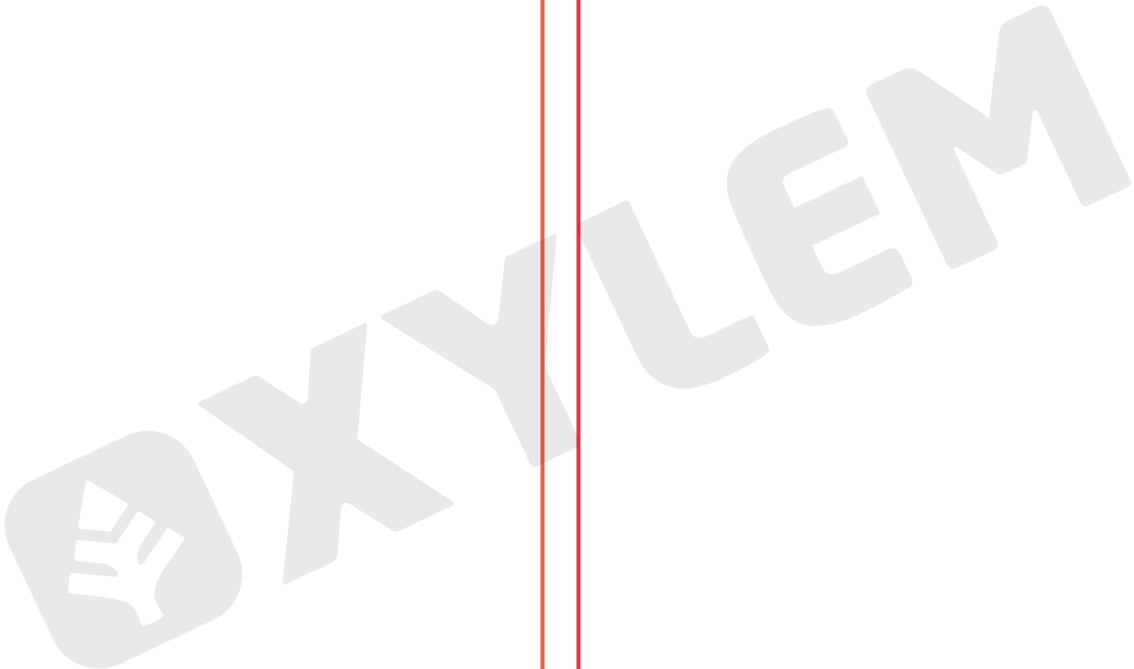
(12) (D)

(13) (D)

(14) (B)

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